

Explainable Fraud Alert Prioritization and Investigation Control Tower

JPMorganChase Inspired Capstone Project Kit

Project domain	Fraud analytics, risk analytics, ML operations, responsible AI
Difficulty	High
Recommended duration	15 calendar days
Package contents	Project brief, dataset, notebook, SQL, KPI, rubric, presentation assets

Direct-to-student execution package. Designed for working professionals and premium portfolio outcomes.

1. Business context

This project converts the bank's public AI-at-scale signals into a realistic fraud operations use case. Students work as part of a fraud analytics team that must detect suspicious transactions, prioritize scarce investigator capacity, and show why alerts were raised.

2. Problem statement

Fraud teams face rising alert volumes, false positives, synthetic identities, and limited investigation bandwidth. The operational challenge is not just to flag risk, but to rank the right cases, explain them clearly, and monitor queue performance.

3. Project objective

Build an explainable fraud-alert triage system that scores transaction risk, prioritizes investigation queues, and provides an investigator-facing control tower with operational metrics and rationale.

4. Project scope

- Use transaction-level data to train or benchmark a fraud detection model.
- Engineer behavioral, velocity, merchant, geo, device, and account-risk features.
- Create alert prioritization tiers and queue assignment logic.
- Add explainability outputs for high-risk alerts.
- Design an investigator dashboard that tracks queue size, hit rate, false positives, and turnaround time.
- Document fairness, explainability, controls, thresholding, and production monitoring assumptions.

5. Expected business outcome

A risk operations manager should be able to reduce missed fraud and wasted investigator time by focusing the queue on the highest-value alerts while preserving transparency and control.

6. What the student must produce

- Fraud modeling notebook
- Investigator dashboard or case-review prototype
- Model card and governance summary
- Threshold strategy memo
- Executive presentation with business case

7. Recommended tools and technologies

- Python
- SQL
- scikit-learn, XGBoost, or LightGBM
- SHAP or comparable explainability technique
- Power BI or Tableau
- Optional Streamlit case-review app

8. Suggested execution workflow

- Define the fraud taxonomy, alert lifecycle, and operating KPIs for investigators.

- Profile class imbalance, label quality, and suspicious behavior patterns in the transaction data.
- Engineer decision-useful features and compare baseline versus advanced classifiers.
- Design triage thresholds that balance precision, recall, and queue size.
- Build an investigator control tower and sample case views with rationale fields.
- Write a governance note that covers explainability, bias, monitoring, and escalation design.

9. Dataset package and file usage guide

This package includes a complete dataset folder, documentation, and supporting assets so students can begin immediately with minimal ambiguity.

Folder	Purpose
B_Dataset/raw	Primary source data for the capstone
B_Dataset/processed	Optional seed outputs to accelerate analysis
B_Dataset/metadata	Data dictionary, schema note, and synthetic data note
C_Supporting_Files	Notebook, SQL, KPI, templates, and submission guidance

Suggested starting order

- Read the project brief before opening the raw data.
- Review the data dictionary and schema note to confirm the business grain.
- Use the starter notebook and SQL file to create your transformed working tables.
- Map every deliverable to the evaluation rubric.

10. Suggested 15-day working plan

Time block	Focus
Days 1-2	Fraud process framing and KPI definition
Days 3-5	Data preparation and feature engineering
Days 6-8	Model development, tuning, threshold testing
Days 9-10	Explainability layer and alert triage logic
Days 11-12	Dashboard or case-review prototype
Days 13-15	Governance note, business case, final presentation

11. Evaluation rubric

Criterion	Weight (%)
Fraud relevance and feature engineering	25
Model evaluation and threshold discipline	25
Explainability and governance quality	15

Operational dashboard usefulness	20
Business case and communication	15

12. Submission instructions

- Submit one executive presentation, one technical workbook or notebook, one concise business report, and one dashboard or prototype artifact.
- Document assumptions, metric definitions, and transformation logic clearly.
- Use business language when presenting trade-offs, not just technical metrics.
- Name all submitted files professionally and include a short reviewer read-me.

13. Interview and CV framing guidance

Strong fit for fraud, banking analytics, fintech, and applied ML roles because it blends model building with operational controls, explainability, and queue design.

Suggested resume framing:

Built a JPMorganChase-inspired analytics capstone that solved a real business problem in fraud analytics, risk analytics, ml operations, responsible ai, combining business framing, analytics execution, and executive communication.

14. Optional advanced extensions

- Add cost-sensitive threshold optimization using estimated fraud loss and investigation cost.
- Segment performance by channel, merchant type, or customer cohort.
- Simulate investigator staffing scenarios or service-level agreements.
- Create a champion-challenger monitoring template for production.

15. Public initiative grounding

JPMorganChase highlighted rapid LLM Suite adoption in 2025 and continues to present AI as a firmwide capability built on secure internal platforms.

The firm's public technology materials and AI research references position fraud detection, behavioral analysis, synthetic data, and AI-enabled operations as sustained areas of value creation.

Reference sources used for capstone grounding

- LLM Suite named 2025 Innovation of the Year by American Banker - JPMorganChase Technology News, June 2025
- Omni means 'all' - JPMorganChase Technology, platform scale-up story
- JPMorgan AI Research pages on Synthetic Data and Time Series & Behavior Analysis - JPMorganChase AI Research